Who is on the other side of your trade?

How exchange rules distribute trading costs and risks, and what this means for protecting everyday traders

Rent or toxicity: strategic exploitability of learning market makers with public wallet information

Deepu Pradeep Miguel Betran Menz

October 27, 2026

6.S974 Games, Learning, and Security, Workshop 1
Example 4: an algorithmic market with adaptive agents

Leverage magnifies losses; liquidation is a future extension

You put in $1,000. At $20\times$ leverage you control a $20,000 position.



collateral: $1,000 **exposure beyond collateral**: $19,000

price $+5\%$

position gain: $20,000 $\times$ 5%



you doubled your money: $+$1,000

price -5%

position loss: $20,000 $\times$ 5%



loss equals collateral: $-$1,000

illustrative PnL before costs; maintenance margin can trigger liquidation before this point

Current model: quotes, fills, information and queue position. It does not track leveraged accounts.

Later extension: collateral, margin and forced sales, to test protections against liquidation losses.

2026: the year retail leverage went wrong at national scale

−44%

KOSPI, June 19 to July 29

1.2 million

retail accounts margin-called: 3.4% of all adults

$39 billion

retail losses on single-stock leveraged ETFs

“Retail” means everyday people

- 92% of the holders were individuals, not firms
- 62% of the accounts: people aged 20 to 39
- taking leveraged exposure through products whose risks can be hard to assess

President Lee Jae-myung, on X, October 14, 2026

살아야지요. 어떻게든 사셔야 합니다. 힘드시면 109번 전화를 하십시오.

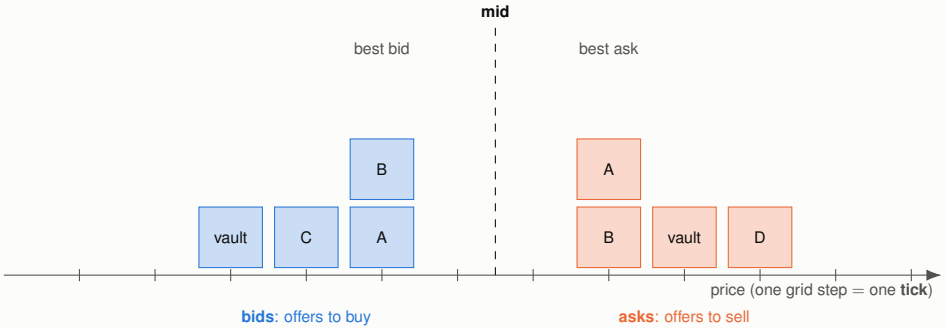
사실 길, 살아야 하는 이유를 함께 찾아드릴 것입니다.

“You must live. Somehow, you must live. If it is hard, call 109. We will find, together, a way and a reason to live.”

Our goal: help traders avoid preventable losses by testing which exchange rules improve their outcomes.

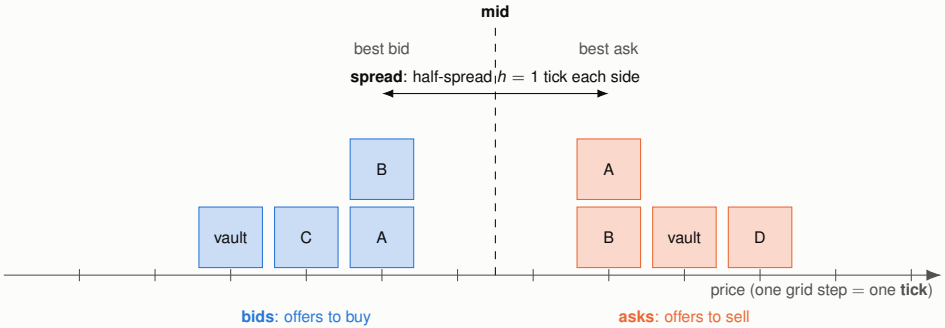
Sources for background figures: Bloomberg, Reuters, Goldman Sachs, Citi (via Seoulz, “KOSPI Crash 2026”); Maeil Shinmun, Oct 22, 2026.

An exchange is a list of standing offers: the order book



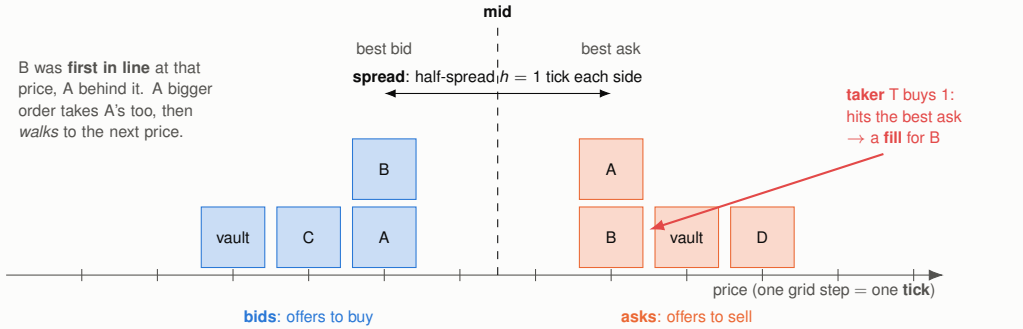
Makers post the offers (the boxes). **Takers** accept them.
Each box carries a public **wallet** address.

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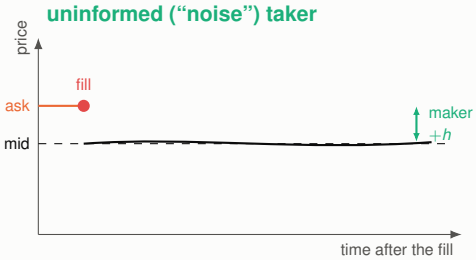
An exchange is a list of standing offers: the order book



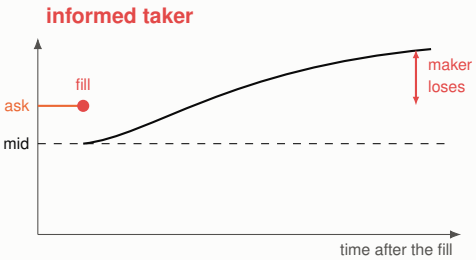
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Public, at once: buy, price, wallet T, wallet B.
Public, only later: whether B gained or lost on it.

Did the maker win? Look at the price afterwards: the markout



price does not move; the maker keeps the half-spread

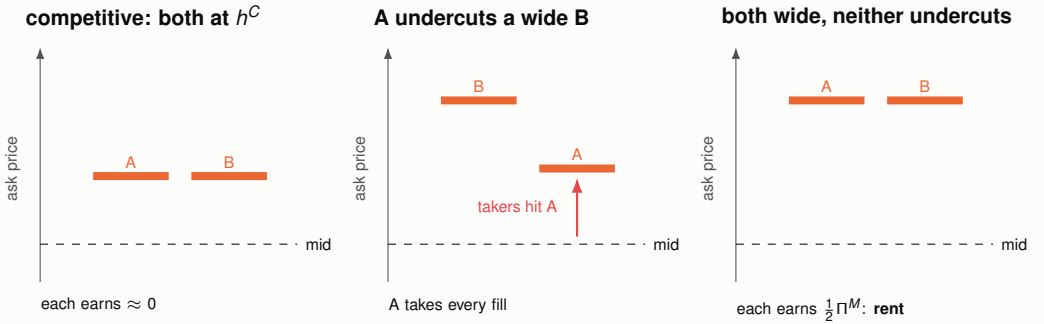


the taker knew; the price runs past the fill price

markout = mid a few seconds later − fill price, signed for the maker

toxic flow: a stream of fills with negative markouts (*adverse selection*)

Rent: what makers earn by not undercutting each other



h^C: the **competitive** quote, break-even. Wider pays, but invites undercutting.

Learning makers end up in the right panel *without talking*: **tacit collusion**. The extra is the **rent**.

A worked example (1): two makers, where the rent comes from, and the price war

Toy payoff table: expected profit per period of the maker with the best quote; one tick $= \$1$

half-spread h	1	2	3	4	5
calm ($\alpha = 0.1$)	0.3	0.8	**1.0**	0.9	0.7
ours ($\alpha = 0.3$)	-0.5	**0**	0.6	**0.9**	0.7
toxic ($\alpha = 0.5$)	-1.5	-0.8	-0.2	0.1	**0.3**

- $h^C = 2$: break-even, the **Nash** quote
- $h^M = 4$: the **joint-monopoly** quote
- **rent** $= 0.9 - 0 = 0.9$ per period
- the same $h = 4$ is a rent in our market and barely break-even in the toxic one

Periods (A, B: the cartel; E: an entrant using the "undercut by one tick" rule)

t	A	B	E	best	profit this period	
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2	4	4	**3**	E	E 0.6, A 0, B 0	

Period 2: undercutting a wide quote takes every fill.

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3	**3**	**3**	**2**	E	E 0, A 0, B 0	cartel matched, E undercut again

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4	2	2	1	E	E −0.5, A 0, B 0	
5	2	2	2	all tie	0, 0, 0	everyone at Nash

Period 2: undercutting a wide quote takes every fill.

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Periods 4–5: the price war ends at Nash. The undercut rule earned 0.6 once and then nothing.

A worked example (2): knowing the reaction, and knowing the wallet

The cartel's rule: match an undercut; return to 4 once nobody undercuts.

An entrant that *knows the rule* never starts the war:

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3	3	3	**5**	A, B tie	E 0, A 0.3, B 0.3	E steps aside
4	**4**	**4**	**3**	E	E 0.6, A 0, B 0	cartel went back to 4
5	3	3	5	A, B tie	E 0, A 0.3, B 0.3	

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per period, on average					E 0.3; A, B 0.15 each (was 0.45)	

Timing the undercut to the cartel's own rule takes a third of the rent here. In the full model the exact best response solves the learned rule by backward induction and takes *all* of it.

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Now add identity. Period 6: E is due to quote 3. The arriving wallet T is one that has cost E money before.

the entrant sees…	A	B	E	profit this period
the **wallet**: T is toxic, step aside	4	4	5	A takes the fill: A −4, E 0
only the **tape**: nothing yet	4	4	3	E takes the fill: E −4

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One toxic fill costs seven good periods. The tape reveals that T was informed only when the price moves, a few fills later; the wallet reveals it *now*. That gap is what identity is worth.

Three players, and what each one gets to know

incumbent makers
learned a cartel; decide *how wide*, and whether to stay in line

an entrant
decides which quotes to undercut, which fills to be there for

the venue
decides the rules: public identities? tick size? pay for position?

the entrant's information set is the *treatment*



What we want to learn

RQ1 Value of identity

What is a public wallet worth to an entrant, beyond the tape, and through which mechanism?

RQ2 Exploitability

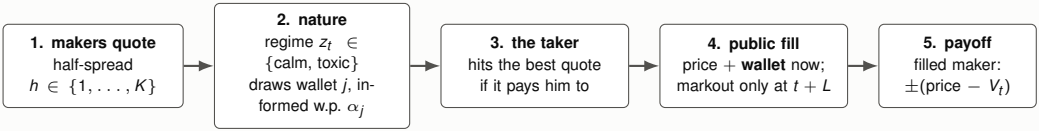
How much of a learned cartel's rent can an adversary take or destroy?

RQ3 One-tick books

When the grid pins the spread, where does rent live, and which rules decide who holds the toxic flow?

Today: identify mechanisms. Next: test rules by customer outcomes.

One period of the game



Incumbents: tabular Q-learners, state = (own last h, rival's last h); trained, then **frozen**.

Entrant: information set $\mathcal{F}^{\text{anon}} \subset \mathcal{F}^{\text{id}} \subset \mathcal{F}^{\text{oracle}}$; a Bayesian filter estimates the next taker's α.

Payoff curve, closed form:
$\Pi(h; \alpha) = (1 - \alpha)\, 2x\bar{\Phi}(x/\sigma_L) - \alpha\mu e^{-x/\mu}$,
$x = h\theta$.

The prototype, in one command

`python experiments/demo.py`

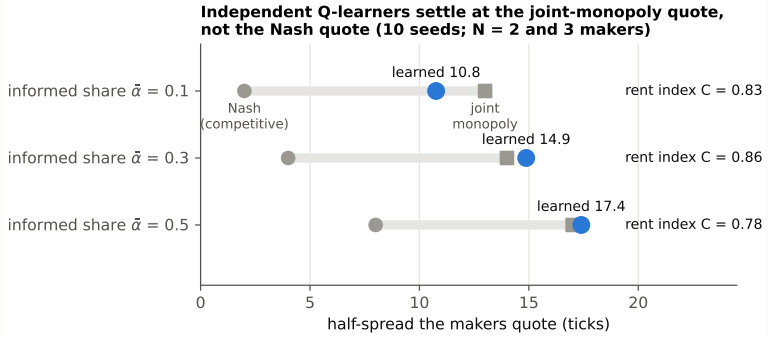
1. benchmarks h^C, h^M, $\Pi(h)$
2. the shipped cartel alone: $C = 0.86$
3. three adversaries enter
4. the best response, period by period

`github.com/pradeepu191/wargame`
59 tests; every result logged with its command.

```
Entry (200 episodes each, same random numbers):
  adversary                          cartel loses  entrant earns  kappa_A
  undercut (one tick inside)              0.41           0.33      0.48
  markout (undercut if rent inferred)     0.47           0.41      0.55
  exact best response                     0.94           1.09      1.10

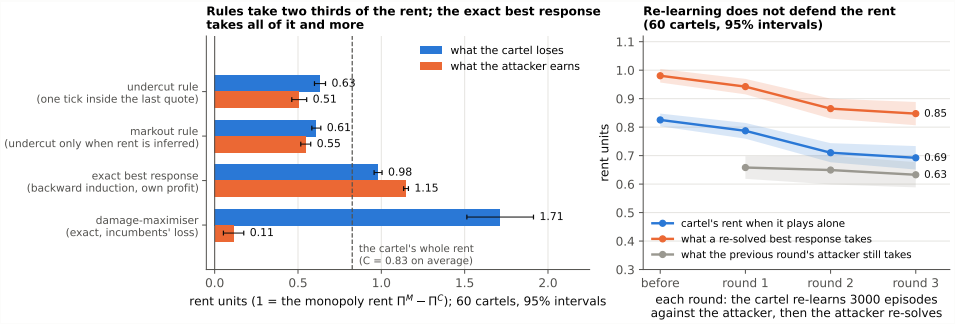
Trace: exact best response vs the cartel (quotes in ticks)
   t  inc0  inc1  entrant   taker          filled   payoff
   0    15    24       12   noise    buy  entrant   +1.21
   1    23    16       13   noise    buy  entrant   +7.86
   2    20    14       13   no trade
   5    21    19       14   informed buy  entrant   -5.77
   7    18    23       14   noise    buy  entrant  +11.79
```

Evidence 1: learners coordinate above the Nash point



$C = 0.71$ to 0.87 in every cell. In 55 to 83% of states undercutting would pay *this period*; the learner forgoes it 96 to 100% of the time.

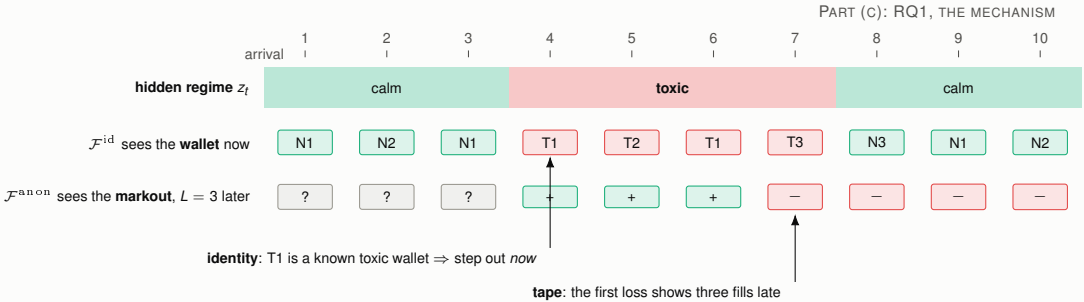
Evidence 2: the rent is fully exploitable



Hand-written rules take two thirds. The **exact best response** takes it all, in 60 of 60 cartels: $\kappa_A = 1.1$ to 1.3.

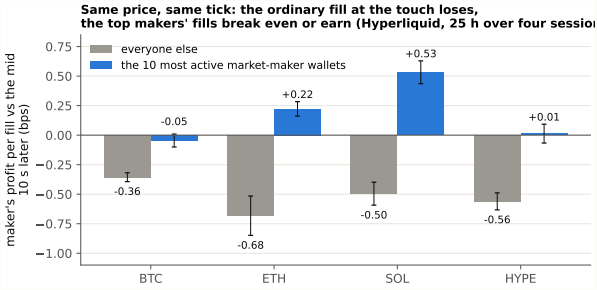
The damage-maximiser destroys 2 to 3$\times$ the rent at no cost.

Why identity has value: a spell of toxic flow, seen two ways



Identity is worth what you lose in the L fills before the tape catches up: measured, zero at $\bar{\alpha} \leq 0.3$, growing with L at $\bar{\alpha} = 0.5$ (backup).

Evidence 3, Hyperliquid: the price cannot move, and someone still earns

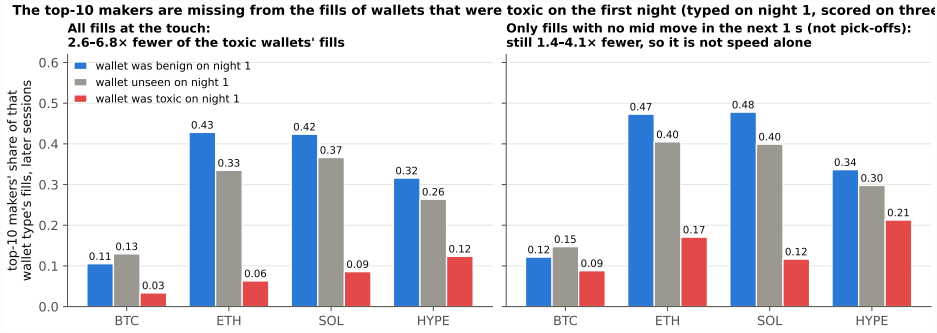


Four sessions, 25 h, 490k scored fills.

- **BTC/ETH/SOL spreads**: essentially one tick
- **Average maker markout**: negative at 10 s on every coin
- **The top makers earn** $+0.3$ to $+1.0$ bps more, *at the same price*; rank persists (corr. 0.5 to 0.8)

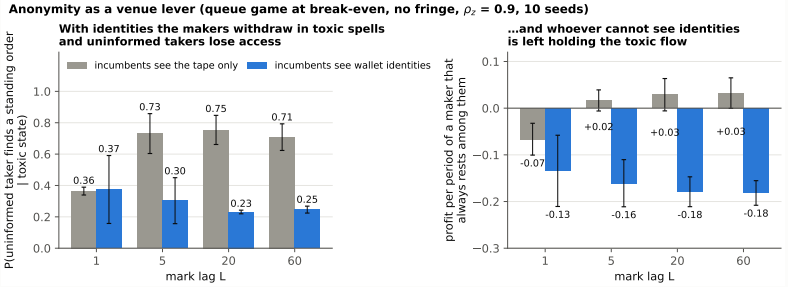
Fill selection matters even at the same quoted price.

Evidence 4: top makers receive fewer toxic-wallet fills



Wallets typed on night 1, scored on three later sessions. Consistent with selection; the makers' decision signals remain unobserved.

Evidence 5: venue rules redistribute access and losses



Anonymity: more uninformed takers get filled in stress in the queue model.

Priority fee: venue revenue rises; the always-resting fringe's losses deepen.

Next test: customer costs and severe losses, alongside access and maker viability.

Workshop 2: test protections for less-informed traders

1 Customer outcomes

Separate small **takers** and **passive traders**: execution costs, gains from trade, stress access and worst-5% losses.

2 Compare rules

Public, anonymous or delayed wallet disclosure. FIFO versus paid priority. Measure **who gains and who loses**.

3 Stress-test benefits

Let makers adapt. Use the existing attacker as a stress test. Check liquidity and whether loss-making participants leave.

Your view: which rule improves less-informed traders' outcomes, and does the improvement survive stress and adaptation?

If you remember three things

1 Vulnerabilities

Learned maker profits are exploitable in our model. This gives us a way to stress-test market rules.

2 Distribution

Information and priority change who gets fills and who bears losses. The data motivates testing these mechanisms.

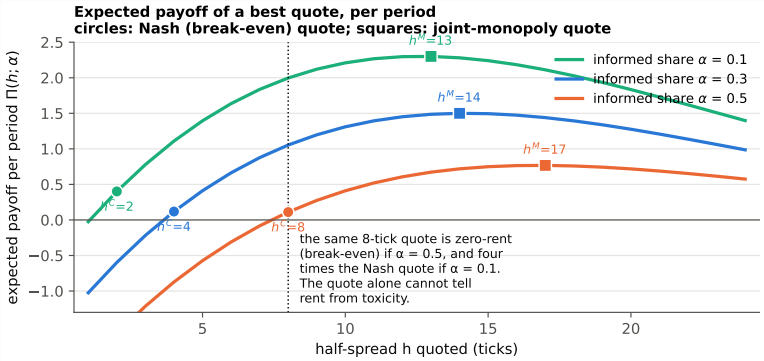
3 Protection next

Test rules by customer costs, access and severe losses. Measure whether less-informed traders benefit.

Goal: fewer preventable losses for everyday traders.

`github.com/pradeepu191/wargame`

Backup: the payoff curve, rent and toxicity look identical



A wide quote is a **rent** . . . or the correct charge for **toxic flow**.

Which curve you are on is decided by α, the share of informed takers. Only outcomes, or *who* trades, reveal it.

Backup: benchmarks are closed-form, so “rent” is measured, not estimated

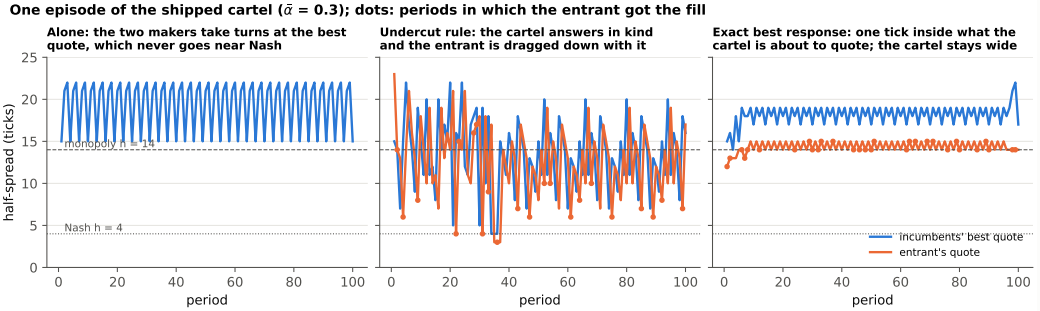
$h^C = \min\{h : \Pi(h) \geq 0\}$	**Nash** (competitive)
$h^M = \arg\max_h \Pi(h)$	**joint monopoly**
$C = \dfrac{\Pi_{\text{cartel}} - \Pi^C}{\Pi^M - \Pi^C} \in [0,1]$	**rent index**
$L_{\mathrm{BR}} = \Pi_{\text{cartel}} - \Pi_{\text{under attack}}$	**loss under attack**
$\kappa_A = L_{\mathrm{BR}}/(\Pi_{\text{cartel}} - \Pi^C)$	**rent destroyed**

At $\bar{\alpha} = 0.3$: $h^C = 4$, $h^M = 14$, rent $\Pi^M - \Pi^C = 1.38$.

Why a cartel is attackable: it is a *reaction function* that does not chase every undercut. That restraint is the rent, and the attack surface.

One-tick (queue) mode: spread pinned; actions REST / OUT / REST-AND-PAY, first in first out. A fill at h^C: $+1.10$ calm, -2.26 toxic.

Backup: steering, period by period



Left: the cartel alone rotates at wide quotes. **Middle**: a naive undercut starts a price war. **Right**: the best response holds the cartel wide and takes the fills.

Backup: longer term, connect execution risk to liquidation losses

Add to the model

- persistent prices and finite book depth
- account collateral and maintenance margin
- forced sales and their price impact

Compare: leverage limits, margin buffers and liquidation execution rules.

Measure protection directly

- fraction of small accounts liquidated
- collateral remaining after liquidation
- severe losses and ability to exit in stress

Evidence: longer recording windows and stress episodes to calibrate the extension.

Reducing liquidation losses is a future hypothesis. Today's model cannot establish it.

Backup: from trading mechanisms to protections we can test

What this study establishes

- learned maker profits can be vulnerable to a strategic entrant
- wallet information can help select fills in the model
- identity and priority rules change who bears toxic flow

The missing test

Does changing a rule improve outcomes for traders with less information and fewer resources?

Public wallets make the question measurable

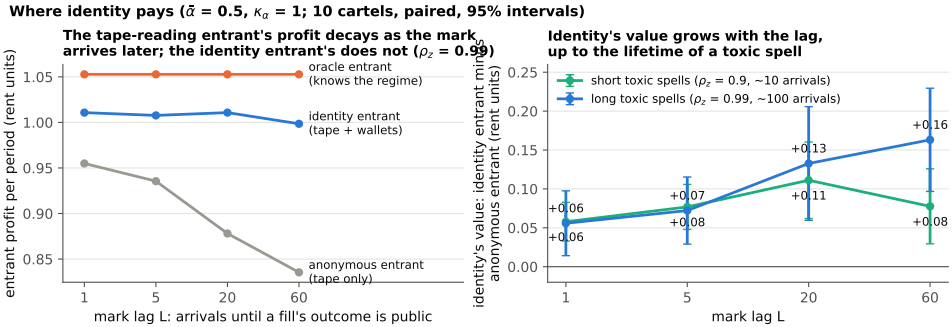
fill buy 0.5 BTC @ 67,412.0
taker `0x7f3a...c91e` maker `0x21b0...4d7a`
a wallet's history is observable

Next: measure customer outcomes

- execution costs and gains from trade
- access to liquidity during stress
- severe losses for passive traders
- whether improvements survive adaptation

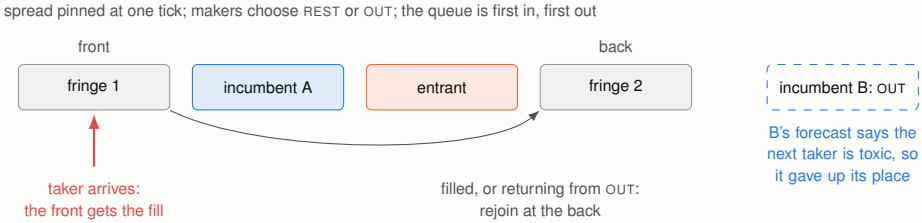
Consumer protection is the next test. Current evidence identifies mechanisms and tradeoffs.

Backup: what identity is worth, measured



The proposal's mechanism (the same wallet trades again) is worth nothing: repeats are 2 to 5%. The lag mechanism survives: zero where $\bar{\alpha} \leq 0.3$, growing with L up to the spell's lifetime.

Backup: the queue game, the one-tick book



A fill at h^C: $+1.10$ in a calm spell, -2.26 in a toxic one.

Who pays in the model: the always-resting *fringe*. Its retail interpretation needs testing.

Backup: Hyperliquid numbers

	BTC	ETH	SOL	HYPE
one tick (bps)	0.12	0.39	0.86	0.11
taker's cost vs the mid: fills at the touch / walked (bps)	0.12 / 2.35	0.42 / 3.20	0.92 / 5.20	0.17 / 3.57
10-s maker markout, at the touch / walked (bps)	-0.41 / -0.80	-0.56 / -2.26	-0.27 / $+0.98$	-0.45 / -0.01
next fill's markout after a toxic vs a benign wallet (bps, 327 h)	-0.64	-1.01	-0.59	-0.59
10-s markout at the touch: top-10 makers / everyone else (bps)	-0.05 / -0.36	$+0.22$ / -0.68	$+0.53$ / -0.50	$+0.01$ / -0.56
per-maker rank correlation, first session vs later (n makers)	0.80 (113)	0.53 (80)	0.70 (53)	0.59 (70)
top-10 share of touch fills: benign / unseen / toxic wallets	.11 / .13 / .03	.43 / .34 / .06	.42 / .37 / .09	.32 / .26 / .12
. . . on fills with no mid move in 1 s: benign / toxic	.12 / .09	.47 / .17	.48 / .12	.34 / .21

Four book sessions over two weeks (25 h). Standard errors cluster by five-minute block; every top-10-vs-rest gap exceeds four SEs. Wallet types from the first night, scored on the three later sessions.

Backup: the double oracle and robustness, in numbers

- **Double oracle** (cartel re-learns 3000 episodes against the fixed best response, entrant re-solves; 3 rounds): the now-stale attacker's profit falls 1.2 $\rightarrow$ 0.5 (the undercut-rule level); the re-solved one takes $\geq$ 90% of the remaining rent in every cartel every round; the cartel's own rent erodes 0.83 $\rightarrow$ 0.79 $\rightarrow$ 0.71 $\rightarrow$ 0.69.
- **Damage-maximiser** ($\lambda = 0$): steers the cartel to 4 to 7 ticks, inside break-even ($h^C = 8$ to 13), by quoting above it; takes no fills, earns ≈ 0; $L_{\mathrm{BR}} = 1.3$ ($\bar{\alpha} = 0.3$), 2.7 to 2.8 ($\bar{\alpha} = 0.5$).
- **Train at $\bar{\alpha}$, test at $\bar{\alpha}'$**: cartels keep $C = 0.80$ to 0.86, $\kappa_A = 0.65$ to 0.75 (rules) on calmer markets; on a more toxic one the rent collapses ($C = 0.23$) and is more than destroyed.
- **Held-out adversaries** preserve every defence ordering (rank correlation 0.75 to 0.96).
- **Caveat we own:** the exact best response needs the cartel's reaction function (oracle knowledge of the *policy*). It is the true upper bound; the public-information attacker that identifies it is next.

Backup: the queue game and the priority fee

- Learned incumbents need a **fringe** of always-resting makers (else a toxic spell empties the book for good); tabular Q cannot learn to queue (time priority pays only after consecutive periods in line); **no-regret (EXP3)** incumbents over complete policies learn to select in a few hundred episodes.
- Against selecting incumbents identity is worth *more*: $+0.05$ to $+0.06$ per period at every lag, 5/5 seeds; the fringe's payoff falls to -0.02 to -0.10: identity-blind resting liquidity is the counterparty of the selection rent.
- **Priority fee**: every selector pays in calm periods (identity-aware incumbents 93 to 94% of calm, 8 to 9% of toxic periods); a non-paying entrant among payers gets 0% of fills; the fringe's loss triples at any fee; revenue peaks at $f = 0.5$.
- **Makers' levers** (20 cartels, 7 adversaries incl. a Q-learning one): requoting every 2 to 5 periods costs everything; randomising costs profit and buys nothing; the identity floor: worst-case profit 0.08 $\rightarrow$ 0.19 at $\bar{\alpha} = 0.5$.